AI Co-Pilot: Combined Aviation Dataset Pipeline

1. RAW AVIATION DATA SOURCES

NASA ASRS Incident Database
6,978 cleaned reports

OpenSky Squawk 7700
832 cleaned emergency trajectories

Normal Flight Envelope Generator
1,716 baseline sequences

2. INGESTION & CONTEXT MIXING (dataset_generator.py)

map_asrs_record()
Phase, severity, hazard and SOP

map_opensky_record()
Call-sign, aircraft type and transponder state

12-Parameter Feature Synthesizer
Altitude, speed, vertical rate, pitch, roll, heading, N1, G, distance, wind, energy and flight phase

3. UNIFIED COMBINED DATASET

9,526 sequences | 15 timesteps per sequence | 12 features per timestep

4. TRAIN / TEST PARTITION (80:20)

Training Set (80%)
7,620 sequential samples

Testing Set (20%)
1,906 sequential samples

5. DATASET COMPOSITION

Total: 9,526 sequences